Project Proposal – AI Content Automation

Project: AI content automation.
Objective: Build an n8n-based system that publishes one short video from Google Drive to YouTube, Instagram, and Facebook with no manual uploading, then collects each platform's analytics after 24–72 hours and turns them into AI-written insights that guide the next posts.

Description :

Content creators spend hours uploading the same short video to many platforms, then guess what worked. This project removes both problems with two connected n8n workflows.

1. **Publishing pipeline.** The post data (video name, title, description, hashtags, platforms, status) starts in a Google Sheet. n8n reads each new row, downloads the matching video from Google Drive and posts it to YouTube, Instagram, and Facebook.

2. **Insights engine.** 24–72 hours after each post, n8n pulls views, likes, comments, shares and watch time from every platform and stores them in one place.

3. **AI analysis.** The combined numbers go to an AI model through OpenRouter (Gemini), which returns a short report: which platform performed best, which posts underperformed, and what to change next time.

The result is a single "upload once, publish everywhere, learn automatically" loop that a solo creator can run for free or at very low cost.

Tools & Design

The system is built as a left-to-right pipeline with a feedback loop at the end:

1. **Source data:** a Google Sheet holds one row per planned post: video file name or Drive link, title, caption, hashtags, target platforms and status.
2. **Trigger:** n8n reads the rows marked "Ready" (on a schedule or when a new row is added).
3. **Video fetch:** n8n finds the matching video in Google Drive, downloads it and validates it (length, size, format).
4. **Publishing:** parallel branches post to YouTube, Instagram and Facebook; each branch writes the post ID, URL and status back to the same sheet row.
5. **Analytics:** a scheduled workflow waits 24–72 hours, then fetches each platform's metrics for every posted row.
6. **AI insights:** the merged metrics are sent to Gemini through OpenRouter, and the report is saved and sent by email or Telegram.

Google Sheet —► n8n Trigger —► Google Drive —► Publish —► Update Log (post data)
(Ready rows) (get video) YouTube (Sheet + move Instagram to Published) Facebook

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▼ Telegram / Gmail ◄— AI Report ◄— OpenRouter ◄— Fetch Analytics (+ save in Sheet)
(insights) (after 24-72h)

Technologies to Use

- **n8n** (self-hosted or cloud) for all workflow automation
- **Business accounts**, a YouTube channel (Brand Account), an Instagram Professional (Business) account linked to a Facebook Page, and a Facebook Page. All are needed to use the publishing and analytics APIs
- **Google Drive API** for video intake and **Google Sheets API** for tracking
- **YouTube Data API v3** and **YouTube Analytics API**
- **Meta Graph API** for Instagram and Facebook publishing and insights
- **OpenRouter API** with Google Gemini for AI insights
- **OAuth 2.0** for platform authorization; **JavaScript** code nodes in n8n for data cleaning
- **Telegram or Gmail** for delivering the insights report

Team Members & Responsibilities

1. **Youssef Ahmed Hassan Elbahrawy** – Team Leader & Project Manager: leads the team, assigns roles and responsibilities, defines the project milestones and weekly plan, tracks progress against deadlines, and coordinates communication between members.
2. **Ebrahim Abdo Elhefnawy Mohammed** – API & Integration Lead: researches and obtains the required API keys and credentials (Google, YouTube, Meta, OpenRouter), sets up the business accounts and authorizations, and builds the initial version of the n8n workflow.
3. **Moustafa Mahmoud Elawady Elsaeed** – Quality & Optimization Lead: reviews the workflow end to end, tests it, fixes errors, improves performance and reliability, and completes any missing parts or steps in the workflow.
4. **Yomna Ahmed Mohammed Hosny** – Documentation & Delivery Lead: prepares the required documents, reports and presentations, and maintains backup copies of the accounts and exported workflows to ensure the project can be recovered and delivered safely.

Week 1: Setup, Design & Intake Workflow

Tasks

- All team members installed n8n, using either Docker or Node.js, and verified that it runs correctly.
- The team started collecting the required API keys and credentials for Google (Drive, Sheets, YouTube), Meta (Instagram and Facebook) and OpenRouter.
- Created the business accounts on YouTube, Instagram and Facebook.

- Prepared the videos required for uploading to the platforms.
- Divided the roles and responsibilities among team members.

Deliverables

- n8n installed and running on all team members' machines
- Business accounts created on all three platforms
- Videos prepared for publishing
- Roles and responsibilities assigned

Week 2: Publishing the first node

Tasks

- Started building the n8n workflow and successfully connected the first node, the YouTube publishing node.
- Team Leader: reviews progress against the milestones, coordinates between members and updates the weekly plan.
- API & Integration Lead: builds the workflow structure, connects the Google Sheets and Google Drive nodes to the YouTube node, and configures the YouTube API authorization (OAuth).
- Quality & Optimization Lead: tests the YouTube node with a sample video from the sheet, documents any errors and fixes them.
- Documentation & Delivery Lead: records the workflow steps completed so far, does a backup of the workflow and accounts, and starts preparing the project documents.

Deliverables

- n8n workflow started, with the YouTube node connected and working
- A test video published to YouTube from the workflow
- Test notes with issues found and fixes
- Backup copy of the current workflow

Week 3: Core Publishing Workflow & Analytics Pipeline

Tasks

- Team Leader: reviews progress against the milestones, coordinates between members and updates the weekly plan.
- API & Integration Lead: builds the Instagram and Facebook publishing branches in parallel with the YouTube branch, reads the post data (title, description, hashtags) from the Google Sheet, and fetches the matching video from Google Drive.
- The whole team collaborates to build the scheduled workflow that fetches YouTube stats (views, likes, comments) and Meta Insights (reach, engagement, shares) and saves the 24-hour and 72-hour snapshots in Google Sheets.

- Quality & Optimization Lead: tests each publishing branch with sample videos, sets up success and error logging in Google Sheets, and moves failed videos to a Failed folder.
- Documentation & Delivery Lead: documents the completed workflow steps, backs up the current workflow and accounts, and continues preparing the project documents.

Deliverables

- Working parallel publishing to YouTube, Instagram and Facebook
- Automatic success/error logging in Google Sheets
- Scheduled analytics workflow storing 24h and 72h performance snapshots
- Updated backup of the workflow

Week 4: AI Insights, Testing & Final Delivery

Tasks

- Team Leader: runs the final review against all milestones, coordinates the final demo and confirms the delivery checklist.
- API & Integration Lead: obtains the free AI API key (OpenRouter free models, or Google AI Studio (Gemini) / Groq as alternatives), integrates the AI node into n8n, and sets up the automated Telegram or Gmail report.
- Quality & Optimization Lead: writes and refines the system prompts for actionable reach recommendations, runs live upload tests across the three platforms, verifies parallel posting reliability and error-handling fallbacks, and tests the AI insights loop with real metrics.
- Documentation & Delivery Lead: completes the user guide, final report and presentation, exports the final workflow files, connects the AI insights report to Telegram or Gmail, configures the automated delivery in n8n, and verifies that the report reaches the recipient correctly.

Deliverables

- Working AI insights engine with automated report delivery (Telegram or Gmail)
- Fully tested end-to-end system: sheet → Drive video → publish → analyze → insights
- Exported n8n workflow files and final backups
- Final documentation, report and presentation